EBA5008: Graduate Certificate in Big Data Analytics

Scalable Epigenomic Data Platform for Biological Age Prediction

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1. Table of Contents

1. Table of Contents **2**
2. Effort Estimates 5
3. Acknowledgements 6
4. Introduction **7**
5. Literature Review 8
Foundational Tissue-Specific and Pan-Tissue Clocks 8
Second-Generation Clocks and Mortality Prediction 8
Technical Infrastructure for High-Throughput Analysis 8
Scalable Cloud Architectures and Automated ML (2025–2026) 8
Longitudinal Validity and Precision (Recent SOTA) 9
6. Business Problems & Objectives **10**
Business Problems 10
Technical Objectives 10
7. Project Design **11**
8. Domain Understanding **12**
Structure of DNA 12
CpG Sites 12
DNA Methylation: Mechanism and Measurement 12
9. Data Understanding **14**
Primary Dataset — PGP UK API and FTP 14
10. Machine Learning - Horvath **15**
Training Data and Study Design 15
Feature Selection: Elastic Net Regression 15
The Age Transformation: Handling Non-Linearity in Development 16
Generating a DNAm Age Prediction 16
Interpreting Outputs: DNAm Age and Epigenetic Age Acceleration 17
11. Machine Learning - Hannum **18**
Training Data and Study Design 18
Feature Selection: LASSO Regression 18
Linear Age Scale 19
Interpreting Outputs: Hannum Age and Age Acceleration 19
12. Machine Learning - GrimAge **21**
How It Works 21
Why It's Significant 21
Practical Application 21
Algorithmic Approach to GrimAge: 22
Stage 1: Predicting "Surrogate" Biomarkers 22
Stage 2: The Mortality Predictor (The "Grim" Part) 22
The Mathematical Logic 22
Why use ML for this? 22

13. Solution Architecture **24**
Architectural Overview 24
Orchestration Design 25
Spark Integration 25
Data Flow Summary 25
14. Cloud Infrastructure Setup - Azure **27**
Resource Group 27
Resource Inventory 28
Network Configuration 29
Virtual Network 29
Network Security Group Rules 29
Azure SQL Firewall Rules 30
Azure SQL Database 30
Resource to Resource Communication 31
Role Based Access Control 32
15. Data Lake **33**
Four Zone Architecture 33
Landing Zone 33
Raw Zone 34
Processing Zone 34
Output Zone 34
Parquet Format Rationale 35
Data Lifecycle Management 35
16. Virtual Machine **36**
VM Specifications 36
Software Installation & Configuration 36
Java and Apache Spark 36
R Packages 36
Python Packages 36
Script Organisation 37
SSH Configuration for Remote Access 37
17. Azure Data Factory **38**
Linked Services 38
Datasets 38
Pipelines 38
load_metadata Pipeline 38
pl_processing Pipeline 39
pl_ml Pipeline 40
Triggers 41
Azure Function App - ADF to VM Bridge 41
SQL Stored Procedures and Design Table 41

18. Power BI Dashboard **43**
Data Connections 43
Visualisations 43
19. Application Layer - Streamlit Application **45**
Individual Report Page 45
Cohort Overview Page 46
IDAT Analysis - Individual 47
Technical Implementation 48
20. Monitoring & Observability 50
Alert Configuration 50
21. Conclusion & Future Scope **52**
Conclusion 52
Future Scope 52
22. References & AI Declaration **53**
AI Declaration 53
References 53

2. Effort Estimates

Name	**Contributions**
Anjali Ancy	Problem Statement Formulation, Machine Learning Algorithm- GrimAge: R & Python Based Testing, Lasso Regression, Cox Mortality Model, Smoking Status Check through Feature Engineering and Support Vector Machine (SVM), Report - 25%, PPT - 20%
Ayaan Mohammed Irfan Khokar	Azure Cloud Infrastructure Setup and Resource Provisioning, Data Lake Zone Architecture, Virtual Machine Configuration and Software Stack Setup (R, Python, Spark), Azure Data Factory Pipeline Development, Azure Function App ADF-to-VM Bridge, Azure SQL Database Design and Stored Procedure Development, Network Security Group Configuration and Firewall Rules Setup, Streamlit Application Development and Deployment, Power BI Dashboard, Monitoring and Observability Setup, Report - 20%, PPT - 25%
Bhavaraju Anuraag	Exploratory Data Analysis, Data Understanding, Domain Understanding, Machine Learning Algorithms - Horvath and Hannum, Report - 20%, PPT - 20%
Lin Yuqin	Data Pipeline, Conclusion and future scope, Report - 15%, PPT - 15%
Reethika Nagineni	Data Lake 4-Zone Architecture Design and Implementation, Network Security Group Configuration,Azure Function App Development and Configuration Azure Monitor and Alerting Setup, Report - 20%, PPT - 20%

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4. Introduction

Biological ageing is a complex, multifactorial process that diverges significantly from chronological age. Two individuals of the same age can exhibit markedly different biological profiles depending on their lifestyle, genetics, environmental exposures, and disease history. Traditional approaches to ageing research have relied on chronological age as a proxy for biological health, which fails to capture the nuanced molecular changes that occur within cells over time. The advent of epigenomics is the study of heritable changes in gene expression that do not involve alterations to the DNA sequence itself and has opened a powerful new avenue for quantifying the biological age of an individual with high precision.

DNA methylation, one of the most well-studied epigenetic marks, involves the addition of a methyl group to cytosine residues at CpG dinucleotide sites across the genome. Methylation patterns at specific CpG sites change predictably with age, forming the molecular basis of so-called 'epigenetic clocks'. Pioneering work by Steve Horvath (2013) demonstrated that a weighted combination of just 353 CpG sites could predict biological age across multiple tissue types with remarkable accuracy. Subsequent clocks including the Hannum clock (calibrated for whole blood) and the GrimAge clock (predictive of mortality risk) have expanded the toolkit available to researchers and clinicians.

Despite the scientific maturity of epigenetic clock methodology, a significant engineering gap exists: processing raw IDAT files from Illumina methylation arrays, computing biological age metrics, and delivering actionable results to researchers at scale remains technically demanding. Existing tools are largely confined to local R environments, lack cloud-native scalability, and do not provide accessible analytical dashboards for non-computational users.

This project addresses these gaps by building NovaGene a fully automated, end-to-end cloud platform on Microsoft Azure that ingests raw epigenomic data from the Personal Genome Project UK (PGP UK), processes 485,512 CpG methylation sites per sample, computes three validated epigenetic clocks (Horvath, Hannum, GrimAge), trains a PySpark-based ML model, and delivers results through both a Power BI cohort dashboard and a Streamlit web application with personalised individual reports. The entire pipeline is orchestrated by Azure Data Factory and monitored via Azure Monitor, making it operationally production-grade despite being built under student subscription constraints.

5. Literature Review

Foundational Tissue-Specific and Pan-Tissue Clocks

The field of epigenetic aging was significantly advanced in 2013 by the work of Hannum et al., who developed one of the first highly accurate DNA methylation (DNAm) clocks. By analyzing whole-blood samples from a diverse cohort, they identified 71 CpG sites that serve as quantitative biomarkers for aging, demonstrating that aging is a genome-wide process that can be captured with high statistical precision. Simultaneously, Horvath (2013) introduced a revolutionary "pan-tissue" clock using 353 CpG sites. This model proved that a single epigenetic ruler could accurately predict the biological age of almost any human tissue or cell type, establishing a universal standard for measuring biological aging and identifying "age acceleration" in diseased tissues.

Second-Generation Clocks and Mortality Prediction

Moving beyond chronological age, **Lu et al. (2019)** introduced **GrimAge**, a "second-generation" clock designed specifically to predict healthspan and mortality. Unlike earlier models, GrimAge was trained on DNAm-based surrogates for smoking history and plasma proteins associated with chronic inflammation and physiological decline. This shift in methodology allowed researchers to better predict the onset of age-related diseases and overall survival, making it a critical tool for clinical trials and longevity research. The model's ability to capture the "biological cost" of lifestyle factors remains a gold standard in the field.

Technical Infrastructure for High-Throughput Analysis

As the volume of genomic data grew, standardized bioinformatics pipelines became essential. **Aryee et al. (2014)** developed the **minfi** package for the Bioconductor project, which provided a comprehensive framework for preprocessing, normalizing, and analyzing Illumina Infinium methylation microarrays. This tool addressed the technical challenges of handling 450K and EPIC array data, enabling researchers to perform quality control at scale. This software foundation is a prerequisite for modern platforms that seek to integrate massive epigenomic datasets from global repositories.

Scalable Cloud Architectures and Automated ML (2025–2026)

Recent research has shifted from individual model development to the creation of end-to-end scalable platforms. We proposed a platform leveraging **Azure Data Factory** and **Spark-based parallel processing** to manage the computational intensity of analyzing nearly half a million CpG sites per sample. By integrating automated machine learning (ML) workflows, the platform demonstrated how cloud-native tools can reduce the time required for DNAm preprocessing and biological age inference. This infrastructure approach allows for real-time interpretation of genomic data, facilitating the transition of epigenetic clocks from research laboratories to large-scale healthcare applications.

Longitudinal Validity and Precision (Recent SOTA)

The most recent state-of-the-art research by **Zhang et al. (2026)** focuses on the longitudinal validity of these clocks. Using the **InCHIANTI cohort**, the study showed that the trajectory of epigenetic age change over time is a more sensitive predictor of mortality than any single-point measurement. This finding emphasizes the importance of longitudinal data tracking and suggests that "third-generation" clocks—which measure the *pace* of aging rather than just accumulated damage—represent the next frontier in precision medicine.

6. Business Problems & Objectives

Business Problems

The field of epigenomic biological age prediction faces three fundamental challenges that limit its real-world applicability:

- Volume: Raw IDAT files from Illumina methylation arrays are large binary files. Processing 485,512 CpG sites per sample across a cohort of 24 samples (yielding 48 IDAT files) requires significant computational resources. Without distributed processing, runtimes become prohibitive as cohort sizes scale.
- Variety: Epigenomic research involves heterogeneous data formats which are binary IDAT files, JSON metadata from REST APIs, ZIP archives from FTP servers, RDS objects from R, and Parquet files from Spark. Integrating these across a unified pipeline without format-induced failures requires careful data architecture design.
- Accessibility: Even when processed, epigenomic results are typically confined to R environments and academic tools inaccessible to clinical researchers, geneticists, or health professionals without computational backgrounds. A user-friendly interface for both cohort-level and individual-level reporting is entirely absent in existing open-source tooling.

Technical Objectives

- Automate weekly data ingestion via Azure Data Factory, pulling from the PGP UK REST API and FTP server into a structured 4-zone Azure Data Lake Gen2.
- Process 485,512 CpG methylation sites per sample using the minfi R package for normalization and SparkR for distributed Parquet output generation.
- Compute three validated epigenetic clocks which are Horvath, GrimAge, and Hannum with age acceleration metrics stored in Azure SQL Database for downstream querying.
- Orchestrate a 5-stage ML pipeline (Validate, Preprocess, Train, Evaluate, Publish) using PySpark RDD-based feature engineering and automated model publishing to GitHub.
- Deliver a Power BI cohort analytics dashboard and a Streamlit web application supporting individual reports with personalised health action plans and real-time IDAT file upload analysis.
- Monitor the entire platform via Azure Monitor with pipeline failure email alerts and execution performance metrics across all compute components.

7. Project Design

Effective project planning and agile execution were critical to successfully navigating the technical complexities of this initiative. Our work was structured using a sprint-based approach, underpinned by rigorous tracking and collaborative tools, ensuring both transparency and momentum.

- **Project Check-ins & Tracking:**

Project progress was continuously monitored. Daily 10–15-minute check-ins addressed immediate blockers and updated next steps, while weekly sprint reviews were used to formally review deliverables and adjust the workload.

- **Clear Accountability:**

We utilized a shared workflow that assigned clear ownership for key processes: Infrastructure Setup, Data Science and ML components, Data Engineering pipelines etc., Roles were occasionally rotated to ensure broad skill development across modules.

- **Tooling and Traceability:**

Our collaboration relied on Teams for real-time communication, Miro for visual planning and initial brainstorming, and version-controlled updates to maintain a transparent and efficient pipeline. All code, data files, and slides were archived for full traceability.

8. Domain Understanding

Structure of DNA

Deoxyribonucleic acid (DNA) is the molecule that carries the genetic instructions for the development, functioning, growth, and reproduction of all known living organisms. It is composed of long chains of nucleotide units. The four bases are adenine (A), thymine (T), guanine (G), and cytosine (C). The sequence in which these bases are arranged along the strand encodes all heritable biological information.

DNA in living cells exists as a double-stranded helix.

It is important to distinguish two levels of genetic information. The genome refers to the complete DNA sequence — the fixed text that is inherited from one's parents and, barring mutations, remains constant throughout life. The epigenome refers to a layer of chemical annotations sitting on top of the genome — a dynamic set of marks that regulate which genes are expressed in which cells and at which times. DNA methylation, discussed in the following subsection, is the best-studied and most stable of these epigenetic marks.

CpG Sites

Across the human genome, cytosine (C) bases that are immediately followed by a guanine (G) on the same strand are referred to as CpG dinucleotides.These sites are of particular biological interest because the cytosine in a CpG context is the primary substrate for a chemical modification known as methylation.

The Illumina Infinium HumanMethylation450 BeadChip (commonly called the 450K array) — the platform on which both the Horvath and Hannum clocks were developed — measures methylation at approximately 450,000 individual CpG sites distributed across the genome. Each site is identified by a unique identifier beginning with 'cg' followed by an eight-digit number (e.g., cg00075967). These identifiers appear throughout the epigenetic ageing literature and are what researchers refer to when discussing specific CpG predictors in a clock model.

DNA Methylation: Mechanism and Measurement

DNA methylation is the addition of a methyl group (CH_3) to the fifth carbon of a cytosine base. The modification does not alter the underlying DNA sequence but it changes how that region of the genome is interpreted by the cellular machinery.

In the context of array-based methylation profiling (the measurement platform used by both Horvath and Hannum), methylation at each CpG site is quantified as a beta value (β), defined as the ratio of methylated signal intensity to total signal intensity:

$$\beta = M / (M + U + \alpha)$$

where M is the fluorescence intensity from the methylated probe, U is the intensity from the unmethylated probe, and α is a small constant (typically 100) added to stabilize estimates at low signal intensities. The resulting value is bounded between 0 and 1:

- A β value of 0 indicates that essentially no copies of the cytosine at that site carry a methyl group (completely unmethylated).
- A β value of 1 indicates that virtually all copies carry a methyl group (completely methylated).
- Intermediate values (e.g., 0.4–0.6) indicate a mixture, which may arise from allele-specific methylation, tissue heterogeneity, or a cell population in which some cells are methylated and others are not.

From a statistical modelling perspective, beta values are a matrix of continuous measurements in [0,1], with rows corresponding to CpG sites (~450,000) and columns corresponding to samples. It is this matrix — often called the methylation matrix or M-matrix — that serves as the input to regression-based epigenetic clock models.

9. Data Understanding

Primary Dataset — PGP UK API and FTP

The primary data source for this project is the Personal Genome Project UK (PGP UK), an open-science initiative that provides public access to participants' genomic, epigenomic, and phenotypic data. Data is accessed via two complementary channels:

- REST API: The PGP UK REST API provides programmatic access to sample metadata in JSON format, including participant characteristics (sex, age, smoking status, tissue type), assay references, and FTP file paths for the corresponding IDAT files.
- FTP Server: Raw IDAT files are hosted on the European Bioinformatics Institute (EBI) FTP server at ftp.ebi.ac.uk. Files are packaged as ZIP archives (one per sample), each containing a pair of IDAT files (_Grn.idat and _Red.idat).

For this project, we ingested 24 samples drawn from the PGP UK cohort, yielding 48 IDAT files with a total of 24 × 485,512 = 11,652,288 individual CpG methylation measurements. Sample metadata includes: chronological age (range: ~30–65 years), sex (male/female), tissue type (blood and saliva), and smoking status (never smoked, smoked occasionally, smoked in the past).

10. Machine Learning - Horvath

The Horvath epigenetic clock, published by Steve Horvath in Nature Genetics in 2013, represents a landmark in the quantitative study of biological ageing. It was the first clock to demonstrate that a single linear model, trained on DNA methylation data, could accurately predict chronological age across a wide variety of human tissues and cell types.

Training Data and Study Design

The model was trained on a large, heterogeneous collection of publicly available methylation datasets. Horvath assembled data from 8,000 samples spanning 51 distinct tissue and cell types, including brain, blood, breast, liver, kidney, lung, saliva, and numerous other tissues. Samples were drawn from multiple independent studies and measured on the Illumina 27K or 450K methylation arrays. Subjects ranged in age from newborns to individuals in their nineties, providing the model with a wide developmental and ageing range.

The pan-tissue design was a deliberate strategic choice. By training simultaneously on multiple tissue types rather than a single tissue, Horvath aimed to identify CpG sites whose relationship with age was stable and consistent regardless of cell type. The resulting clock captures what might be called a universal biological ageing signal — a methylation drift that runs in parallel with chronological time across all cell lineages, and which is therefore thought to reflect an intrinsic cellular ageing process rather than tissue-specific wear.

Feature Selection: Elastic Net Regression

The central statistical challenge faced by Horvath was one of extreme dimensionality: the input data comprised approximately 450,000 methylation features (CpG sites) per sample, while the number of training samples was approximately 8,000. This setting — many more predictors than observations — makes ordinary least squares regression entirely impractical. The coefficient matrix would be underdetermined and the resulting model would overfit massively to the training data.

Horvath addressed this by applying elastic net regression, a penalised regression method that combines two types of regularisation:

- L1 regularisation (LASSO component): adds a penalty equal to the sum of absolute values of all coefficients. This drives many coefficients exactly to zero, effectively performing automatic variable selection.

- L2 regularisation (Ridge component): adds a penalty equal to the sum of squared coefficients. This shrinks remaining coefficients toward zero and handles correlated predictors more gracefully than pure LASSO, preventing the model from arbitrarily selecting one member of a correlated group.

Applied to the methylation matrix, elastic net reduced 450,000 candidate CpG features to a final set of 353 sites with non-zero coefficients. These 353 sites form the clock. Of these, 193 show increasing methylation with age (positive coefficients) and 160 show decreasing methylation with age (negative coefficients). The selected sites are distributed across the genome and span promoters, gene bodies, intergenic regions, and CpG islands.

The Age Transformation: Handling Non-Linearity in Development

A critical and often underappreciated feature of the Horvath clock is its treatment of age as the outcome variable. Methylation does not change at a constant rate across the lifespan. During foetal development and early childhood, methylation patterns change extremely rapidly — the epigenome is being actively remodelled as cells differentiate and mature. After approximately age 20, the rate of methylation change slows considerably and becomes more linear with respect to time.

To capture this non-linearity, Horvath applied a piecewise age transformation before fitting the regression model. Rather than predicting chronological age directly, the model is trained to predict a transformed version of age, defined as:

F(age) = log(age + 1) − log(adult + 1) if age ≤ 20

F(age) = (age − adult) / (adult + 1) if age > 20

where 'adult' is set to 20. The first branch applies a logarithmic compression to ages up to 20 years, reflecting the rapid methylation changes of development. The second branch is a linear scaling for ages above 20, reflecting the more gradual changes of adulthood. After model fitting, predictions are back-transformed to the original age scale using the inverse of these functions.

Generating a DNAm Age Prediction

For a new sample, the Horvath clock generates a predicted biological age through the following sequence of steps:

- Extract the beta values for each of the 353 clock CpG sites from the methylation array data.
- Compute the weighted linear combination of these 353 beta values using the published model coefficients, plus the intercept term.
- Apply the inverse age transformation to convert the model output back to a predicted age in years.

The weighted sum in step 2 takes the form:

$$\textbf{DNAm age (transformed)} = w_0 + w_1\beta_1 + w_2\beta_2 + \ldots + w_{353}\beta_{353}$$

where w_0 is the intercept, w_i are the 353 learned regression coefficients, and β_i are the measured beta values at the corresponding CpG sites. The inverse transformation then yields the final DNAm age estimate in years. Horvath's published coefficients for all 353 sites are available as supplementary data to his 2013 paper and are implemented in widely used R packages such as methylclock and ChAMP.

Interpreting Outputs: DNAm Age and Epigenetic Age Acceleration

The primary output of the Horvath clock is the DNAm age: a predicted biological age in years, derived entirely from the methylation state of 353 CpG sites. In healthy individuals, DNAm age tracks chronological age closely, with a mean absolute error of approximately 3.6 years across tissues. This correspondence is remarkable given that the model was not trained on any phenotypic or health information — only methylation measurements and chronological age.

The more biologically informative quantity is the epigenetic age acceleration, defined as the residual obtained by regressing DNAm age on chronological age:

$$\textbf{Age Acceleration} = \textbf{DNAm Age} - \textbf{Chronological Age}$$

A positive value indicates that the individual's epigenome appears biologically older than their calendar age; a negative value indicates the reverse. Importantly, this residual captures variation in biological ageing that is independent of chronological age — it is not simply that older people have higher age acceleration, but that some individuals age faster or slower than their peers of the same chronological age.

11. Machine Learning - Hannum

The Hannum epigenetic clock, published by Gregory Hannum and colleagues in Molecular Cell in 2013 took a different but complementary approach to epigenetic age estimation. Where Horvath pursued universality across tissues, Hannum focused on a single, clinically accessible tissue (whole blood) and incorporated biological covariates explicitly into the modelling framework. This section describes its design, statistical methodology, and the ways it differs from and complements the Horvath clock.

Training Data and Study Design

The Hannum clock was trained on a cohort of 656 whole-blood samples collected from individuals at the University of California San Diego, spanning ages 19 to 101 years. Blood was chosen because it is the most routinely collected biological material in clinical and epidemiological research, making the resulting clock directly applicable to large-scale population studies and biobank data. All samples were measured on the Illumina Infinium HumanMethylation450 BeadChip array.

The deliberate restriction to a single tissue type is a fundamental design difference from Horvath. By limiting the training data to blood, Hannum accepted reduced generalisability across tissues in exchange for a model that is more tightly calibrated to the methylation dynamics of blood specifically.

Feature Selection: LASSO Regression

Like Horvath, Hannum faced a high-dimensional regression problem: 450,000 methylation features, 656 samples. However, Hannum chose LASSO (Least Absolute Shrinkage and Selection Operator) rather than elastic net as the penalised regression method. LASSO applies only an L1 penalty to the regression coefficients:

Minimise: RSS + $\lambda \sum|w_i|$

The L1 penalty induces sparsity by driving many coefficients to exactly zero. This produces a model that uses only a subset of available features — those whose methylation-age relationship is strong enough to survive the penalty. Applied to the 450K methylation data, LASSO reduced the feature space from approximately 450,000 CpG sites to a final set of 71 predictors.

The sparser model (71 sites vs Horvath's 353) reflects two factors: the use of pure L1 rather than L1+L2 regularisation, and the restriction to a single tissue type. When training data

come from a single tissue, the methylation signals are more homogeneous, and a smaller number of sites suffices to explain the age variance. In a pan-tissue model, more sites are needed to capture the additional variance introduced by tissue differences.

Linear Age Scale

A notable simplification in the Hannum model, relative to Horvath's, is the use of a straightforward linear relationship between methylation and age. Because the Hannum training cohort was restricted to adults (ages 19+), the highly non-linear methylation changes associated with foetal development and childhood are absent from the data. In the adult age range, the relationship between DNA methylation and age is sufficiently linear that no transformation of the outcome variable is required.

The model therefore takes the standard linear regression form:

$$\textbf{Predicted Age} = w_0 + w_1\beta_1 + w_2\beta_2 + \ldots + w_{71}\beta_{71}$$

where w_0 is the intercept, w_i are the 71 LASSO-selected regression coefficients, and β_i are the measured beta values at the corresponding CpG sites. The output is directly in years — no back-transformation is required. This makes the Hannum clock computationally simpler to implement and the coefficients more directly interpretable as the change in predicted age per unit increase in beta value at each site.

Interpreting Outputs: Hannum Age and Age Acceleration

The primary output of the Hannum clock is a predicted biological age in years, derived from the weighted sum of 71 CpG site beta values. In the original validation, the model achieved a Pearson correlation of $r = 0.96$ with chronological age in held-out test samples, with a median absolute deviation of approximately 3.9 years. Performance was consistent across the adult age range.

As with the Horvath clock, the scientifically interesting quantity is not the predicted age itself but the age acceleration residual:

Hannum Age Acceleration = Hannum DNAm Age − Chronological Age

Positive acceleration indicates a biologically older-appearing methylome; negative indicates the reverse. Hannum demonstrated that this residual was associated with sex (women showed lower age acceleration than men on average), BMI (higher BMI correlated with higher acceleration), and blood cell composition. After adjusting for these covariates, the residual

age acceleration was associated with independent markers of health, including cancer status in a subset of samples.

Because the Hannum clock was trained exclusively on blood, it tends to be more sensitive to processes that affect the blood compartment specifically — immune aging, chronic inflammation, and haematopoietic stem cell dysfunction. This makes it particularly well-suited for studies of immune ageing, infectious disease, and conditions that manifest in the haematopoietic system.

12. Machine Learning - GrimAge

GrimAge is a specialized type of "epigenetic clock" used by researchers to estimate biological age and predict lifespan. Developed by Steve Horvath and his team in 2019, it is widely considered one of the most accurate predictors of mortality and age-related health decline.

How It Works

Unlike chronological age (the years since you were born), **GrimAge** measures biological age by looking at:

- **DNA Methylation:** It analyzes specific chemical markers (methyl groups) attached to your DNA. As we age, these markers change in predictable patterns.
- **Plasma Protein Biomarkers:** Most epigenetic clocks only look at DNA, but GrimAge is unique because it was trained to predict the levels of seven specific blood proteins associated with inflammation and heart health (such as GDF15 and cystatin C).
- **Smoking History:** It also incorporates an epigenetic signature for smoking pack-years, which is a significant factor in cellular aging.

Why It's Significant

The "Grim" in the name is a nod to the **Grim Reaper**, reflecting its primary purpose: predicting "time-to-death."

- **Mortality Prediction:** Research has shown that if your GrimAge is higher than your actual chronological age (known as "age acceleration"), you may have a higher risk of developing chronic diseases or a shorter life expectancy.
- **Health Span vs. Life Span:** It is used in clinical trials to see if specific interventions (like diet, exercise, or new medications) can actually "slow down" the clock or even reverse biological aging.
- **Accuracy:** It generally outperforms earlier epigenetic clocks (like the original Horvath Clock) because it links DNA changes directly to physiological health indicators and mortality data rather than just chronological age.

Practical Application

While currently used mostly in **academic research** and **clinical trials**, some longevity-focused health companies now offer GrimAge testing to consumers. These tests allow individuals to see how their lifestyle choices might be impacting their biological rate of aging.

Algorithmic Approach to GrimAge:

GrimAge is calculated using a **two-stage machine learning process** that primarily relies on **Elastic Net Regression**.
Unlike simpler clocks that try to predict chronological age directly from DNA, GrimAge is a "composite" biomarker. It uses ML to bridge the gap between your DNA, your blood chemistry, and your actual risk of mortality.

Stage 1: Predicting "Surrogate" Biomarkers

Instead of jumping straight to age, the model first predicts the levels of specific health-related variables based solely on DNA methylation data (DNAm).

- **The Features:** The input consists of thousands of **CpG sites** (specific locations on your DNA where methyl groups attach).
- **The Targets:** The model is trained to predict the plasma levels of **seven proteins** (like GDF15 and Cystatin C) and **smoking history** (pack-years).
- **The Method: Elastic Net Regression** is used here. It's an ideal choice because it handles "high-dimensional" data (where you have many more features than samples) by performing automatic feature selection—effectively "turning off" CpG sites that don't help predict the protein levels.

Stage 2: The Mortality Predictor (The "Grim" Part)

Once the model has the predicted values for those seven proteins and smoking history, it combines them into a single score.
Cox Regression: The researchers used a **Cox Proportional Hazards Model** to relate these DNAm-based surrogates, along with age and sex, to "time-to-death" data.
The Output: The resulting mortality risk score is then **linearly transformed** back into a unit of "years." This makes the output intuitive—if your GrimAge is 65 but your chronological age is 60, the ML has detected molecular signatures typically seen in someone five years older.

The Mathematical Logic

The final calculation looks roughly like a weighted sum:

$$DNAmGrimAge = \beta_0 + \beta_1(Age) + \beta_2(Sex) + \sum_{i=1}^{8} \omega_i(Surrogate_i)$$

Where:

- Beta and Omega are **weights** learned during the training phase.
- Surrogate_i are the values predicted in Stage 1 (the 7 proteins + smoking).

Why use ML for this?

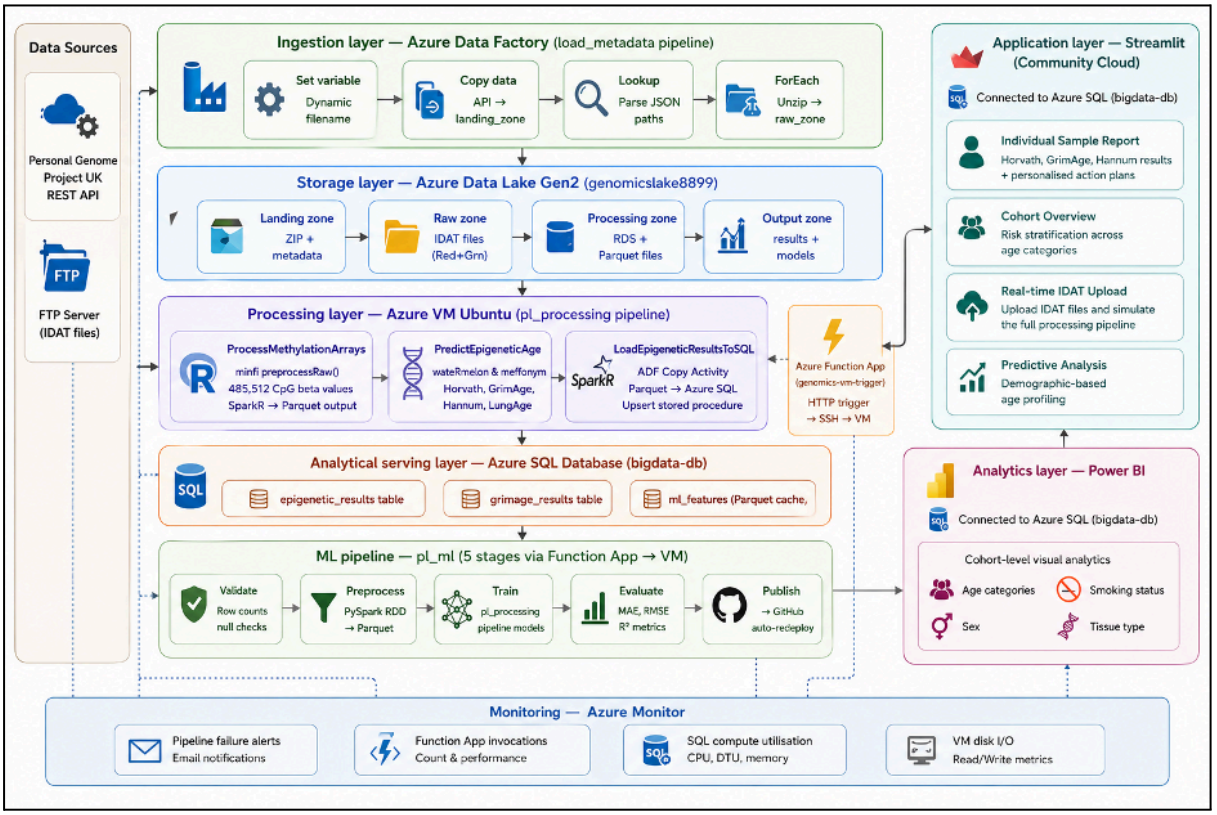
- **Noise Reduction:** DNA methylation data is incredibly "noisy." ML models like Elastic Net act as a filter, identifying the few hundred sites out of 450,000+ that actually matter for health.
- **Surrogate Modeling:** The real genius of GrimAge is using ML to "proxy" blood tests.

Even if you don't have a physical blood sample to test for those seven proteins, the ML model can "see" their impact on your DNA.

- **Non-Linearity:** While the final transformation is linear, the initial selection process accounts for the complex, multi-variable relationships between genetics and lifestyle factors like smoking.

13. Solution Architecture

The NovaGene platform is a cloud-native, end-to-end epigenomic data engineering system designed to ingest, process, analyse, and serve biological age predictions derived from DNA methylation array data. The architecture follows a multi-layered design principle aligned with modern big data engineering best practices, combining distributed storage, orchestrated pipeline execution, Spark-based computation, and a cloud-deployed analytical application



Architectural Overview

The system is built entirely on Microsoft Azure and is structured around six functional layers that operate in sequence:

- Data Sources Layer — The Personal Genome Project UK (PGP UK) REST API and FTP servers serve as the primary external data sources, providing participant metadata and raw IDAT methylation array files respectively. User-uploaded IDAT files from the Streamlit application constitute a secondary real-time data source.
- Ingestion Layer — Azure Data Factory (ADF) orchestrates scheduled ingestion via the load_metadata pipeline, pulling data from the PGP UK API and FTP into the Azure Data Lake landing zone every Monday at 7:00 AM SGT.
- Storage Layer — Azure Blob Storage Gen2, configured as a Data Lake, stores all data across four immutable tiered zones: landing, raw, processing, and output. Outputs are stored in Apache Parquet format to ensure scalability and compatibility with distributed processing frameworks.

- Processing Layer — An Ubuntu 22.04 Virtual Machine hosts all heavy computation including R-based methylation array processing via the minfi package, epigenetic clock predictions via wateRmelon and meffonym, and Spark-based distributed data transformation via SparkR and PySpark.
- Analytical Serving Layer — Azure SQL Database serves as the structured output store, holding computed epigenetic results in three tables: epigenetic_results, grimage_results, and ml_features.
- Application Layer — A Streamlit application deployed on Streamlit Community Cloud serves as the clinical-facing interface, providing individual sample reports, cohort analytics, real-time IDAT analysis, and predictive age profiling. A Power BI dashboard provides aggregated cohort-level visual analytics.

Orchestration Design

Azure Data Factory acts as the single orchestration layer across all pipelines. Three production pipelines are defined: load_metadata for ingestion, pl_processing for methylation processing and age prediction, and pl_ml for the ML feature engineering and model publishing workflow. A critical engineering constraint was that the Azure for Students subscription restricts Entra ID service principal creation. This was resolved by introducing an Azure Function App (genomics-vm-trigger) as an HTTP bridge: ADF Web Activities call the Function App over HTTP, which in turn SSHes into the VM and executes the designated script, returning the exit status to ADF for conditional branching.

The screenshot below shows the complete pipeline canvas in ADF Studio, illustrating the sequential activity flow from ingestion through to the processing and ML pipeline invocations:



Spark Integration

Apache Spark is integrated at two points in the architecture to fulfil the distributed processing requirement. SparkR is used within the pl_processing pipeline to convert processed methylation results into Parquet format before uploading to the Data Lake. PySpark is used in the pl_ml pipeline's preprocessing stage where RDD operations encode and transform features from both SQL tables into a unified Parquet feature set. This ensures that all intermediate and final data artefacts conform to big data storage standards and can be consumed by any Spark-compatible analytical workload.

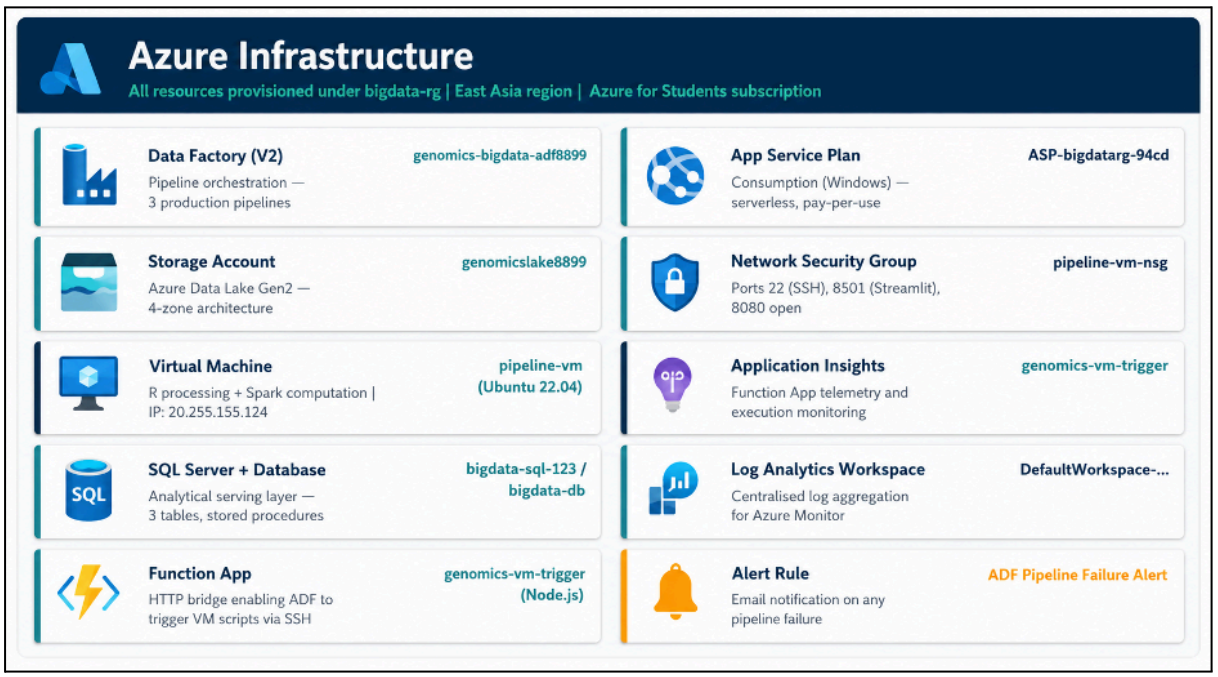
Data Flow Summary

Stage	Data Flow

Ingestion	PGP UK API + FTP → ADF → Data Lake landing_zone + raw_zone
Processing	raw_zone IDAT files → VM (minfi) → processing_zone (RDS + Parquet)
Age Prediction	processing_zone → VM (wateRmelon + meffonym) → output_zone (Parquet)
SQL Loading	output_zone Parquet → ADF Copy Activity → Azure SQL Database
ML Pipeline	Azure SQL → VM (PySpark RDD) → processing_zone → GitHub
Serving	Azure SQL → Streamlit App + Power BI Dashboard

14. Cloud Infrastructure Setup - Azure

The entire cloud infrastructure for the NovaGene platform was provisioned from scratch on a Microsoft Azure for Students subscription. All resources were created and configured manually through the Azure Portal and grouped under a single resource group to maintain logical isolation, cost tracking, and lifecycle management.



Resource Group

A dedicated resource group named bigdata-rg was created in the East Asia region (Azure's Hong Kong data centre). Centralising all resources under a single resource group ensures unified access control, simplified monitoring, and the ability to manage all components as a single deployable unit.

Resource Group	bigdata-rg
Region	East Asia (Hong Kong)
Subscription	Azure for Students

The following screenshots show the complete resource inventory within bigdata-rg:

Name	Type	Location	Resource Group	Subscription	Last accessed
bigdata-rg	Resource group		bigdata-rg	Azure for Students	11 days ago
pipeline-vm-nsg	Network security gro...		bigdata-rg	Azure for Students	13 days ago
master	SQL database		bigdata-rg	Azure for Students	13 days ago
NetworkWatcher_eastasia	Network Watcher		NetworkWatcherRG	Azure for Students	13 days ago
NetworkWatcherRG	Resource group		NetworkWatcherRG	Azure for Students	13 days ago
Azure for Students	Subscription			Azure for Students	13 days ago
pipeline-vm-ip	Public IP address		bigdata-rg	Azure for Students	1 month ago
pipeline-vm636	Network Interface		bigdata-rg	Azure for Students	1 month ago
pipeline-vm-vnet	Virtual network		bigdata-rg	Azure for Students	1 month ago
genomicslake8899	Storage account		bigdata-rg	Azure for Students	7 days ago
pipeline-vm	Virtual machine	East Asia	bigdata-rg	Azure for Students	7 days ago
genomics-vm-trigger	Function App	East Asia	bigdata-rg	Azure for Students	7 days ago
bigdata-db	SQL database	East Asia	bigdata-rg	Azure for Students	7 days ago
genomics-bigdata-adf8899	Data factory (V2)	East Asia	bigdata-rg	Azure for Students	8 days ago
bigdata-sql-123	SQL server	East Asia	bigdata-rg	Azure for Students	8 days ago
MSVMI-eastasia-pipeline-vm	Data collection rule		bigdata-rg	Azure for Students	8 days ago
ADF Pipeline Failure Alert	Metric alert rule		bigdata-rg	Azure for Students	8 days ago
genomics-vm-trigger	Application Insights	East Asia	bigdata-rg	Azure for Students	11 days ago
ASP-bigdatarg-94cd	App Service plan	East Asia	bigdata-rg	Azure for Students	11 days ago
DefaultWorkspace-9b68e170-9bf8-484b-9b...	Log Analytics worksp...	East Asia	DefaultResourceGrou...	Azure for Students	11 days ago

Resource Inventory

The following Azure resources were provisioned as part of the platform infrastructure:

Resource	Name	Type	Purpose
Data Factory	genomics-bigdata-adf8899	ADF V2	Pipeline orchestration
Storage Account	genomicslake8899	Blob Storage Gen2	Data Lake — all zones
Virtual Machine	pipeline-vm	Ubuntu 22.04 LTS	R + Spark computation
SQL Server	bigdata-sql-123	Azure SQL Server	Database hosting
SQL Database	bigdata-db	Serverless Gen5	Analytical serving layer
Function App	genomics-vm-trigger	Consumption Windows	ADF-to-VM SSH bridge
App Service Plan	ASP-bigdatarg-94cd	Consumption Plan	Function App hosting
App Insights	genomics-vm-trigger	Application Insights	Function telemetry

Log Analytics WS	DefaultWorkspace-...	Log Analytics	Centralised log store
Network Watcher	NetworkWatcher_eastasia	Network Watcher	Network diagnostics
Alert Rule	ADF Pipeline Failure Alert	Metric Alert	Pipeline failure emails
NSG	pipeline-vm-nsg	Network Security Group	VM port access control

Network Configuration

Networking was configured to ensure secure communication between all Azure resources while allowing necessary external access for pipeline execution and application serving. The following network rules and configurations were established:

Virtual Network

The VM was provisioned inside a Virtual Network (pipeline-vm-vnet) with a dedicated subnet. This allows resources within the same VNet to communicate over private IP addresses without traversing the public internet. The VM was assigned a static public IP (pipeline-vm-ip: 20.255.155.124) to allow SSH access and Streamlit serving.

Network Security Group Rules

A Network Security Group (pipeline-vm-nsg) was attached to the VM's network interface with the following inbound port rules:

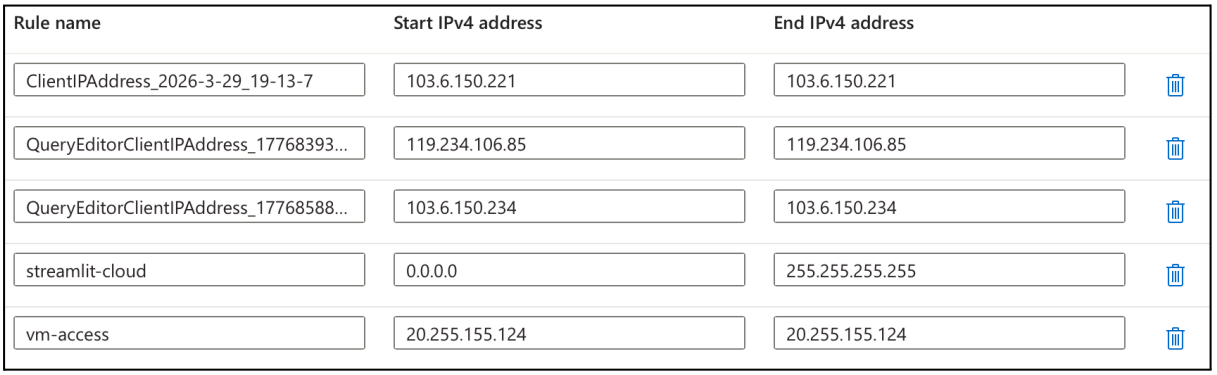
Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (6)						
300	SSH	22	TCP	Any	Any	Allow
310	airflow8080	8080	TCP	Any	Any	Allow
320	streamlit	8501	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Port / Rule	Purpose
Port 22 (TCP) — SSH	Remote access to VM for development, script execution, and maintenance
Port 8501 (TCP) — Streamlit	Exposes Streamlit application to the public internet

Port 8080 (TCP) — Airflow	Reserved port opened for potential Airflow integration
65000 — AllowVnetInBound	Allows all traffic between resources within the same VNet
65001 — AzureLoadBalancer	Allows Azure Load Balancer health probe traffic
65500 — DenyAllInBound	Default deny — all unmatched inbound traffic is blocked

Azure SQL Firewall Rules

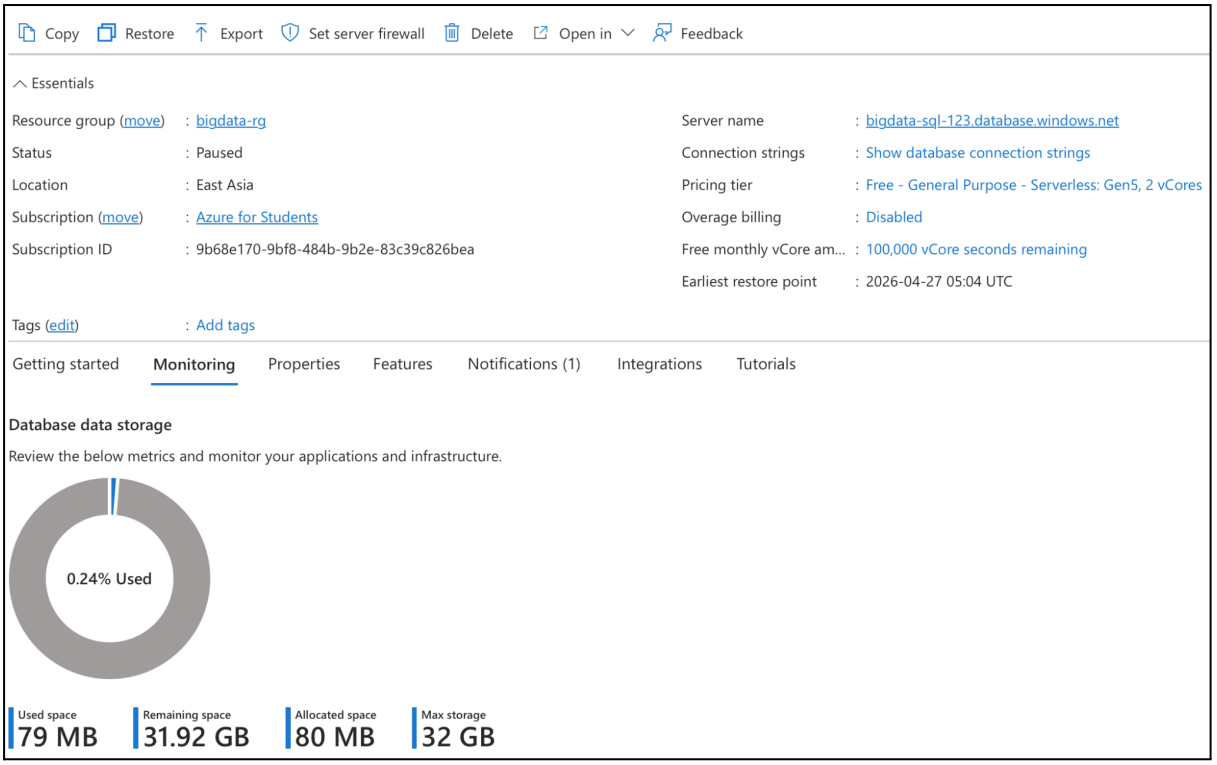
The Azure SQL Server firewall was configured to allow connections from specific IP addresses as well as from all Azure services:



Firewall Rule	Purpose
Allow Azure services	Enables ADF Integration Runtime (East Asia) to connect to SQL for Copy Activities
VM IP (20.255.155.124)	Allows the pipeline VM to connect to SQL for writing ML pipeline outputs
QueryEditorClientIPAddress	Auto-added rules enabling portal Query Editor access from developer machines
0.0.0.0 — 255.255.255.255	Opened for Streamlit Community Cloud access (demo purposes)

Azure SQL Database

The SQL Database bigdata-db was provisioned on the Serverless Gen5 tier with auto-pause enabled. The database overview confirms its Online status and East Asia location:



Resource to Resource Communication

- ADF → Data Lake: ADF uses the AutoResolveIntegrationRuntime in East Asia to connect to Azure Blob Storage Gen2 via the storage account's access key stored in ADF linked services.

- ADF → Function App: ADF Web Activities call the Function App's public HTTPS endpoint using the function's secret key appended as a query parameter (?code=...). No VNet integration is required as both ADF and the Function App communicate over the public HTTPS endpoint.

- Function App → VM: The Function App SSHes into the VM over the public internet using the VM's public IP (20.255.155.124) on port 22. The VM password is stored as an environment variable (VM_PASSWORD) in the Function App's application settings rather than hardcoded in the function code.

- VM → Data Lake: The VM uses the AzureStor R package and azure-storage-blob Python library to upload and download files directly to the Data Lake storage account using the storage account access key.

- VM → Azure SQL: The VM connects to Azure SQL using the pymssql Python library and pyodbc with ODBC Driver 18 for SQL Server, authenticating with SQL Server credentials over port 1433.

- ADF → Azure SQL: ADF uses the AzureSqlDatabase1 linked service with SQL authentication to execute Copy Activities that load Parquet data from the Data Lake into SQL tables.

- Streamlit Cloud → Azure SQL: The Streamlit application deployed on Streamlit Community Cloud connects to Azure SQL using pymssql with credentials stored in Streamlit's secrets management system.

- Power BI → Azure SQL: Power BI connects directly to Azure SQL using the server FQDN and SQL authentication credentials, with scheduled refresh configured for automatic data updates.

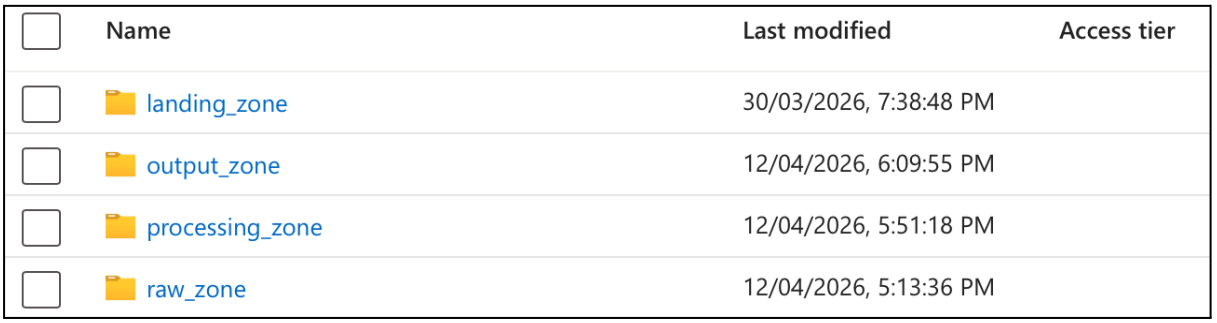
Role Based Access Control

Role-based access was configured to allow all team members to access the Azure resources. The following access structure was established under the bigdata-rg resource group:

- Owner: Ayaan — full access to create, modify, and delete all resources
- Contributor: All team members (Anuraag, Anjali, Reethika, LinYuqin) were added as Contributors to the resource group, granting them read and write access to all resources without the ability to manage role assignments
- SQL access: Each team member was provided with the SQL admin credentials to access the Azure SQL Query Editor and connect from local tools
- VM access: SSH access to the VM was configured for team members — each member's public SSH key or password was added to the VM to allow remote login as genomics-user
- Storage access: The storage account access key was shared with the team for direct blob access where needed.

15. Data Lake

The data storage layer of the NovaGene platform is built on Azure Blob Storage Gen2 with hierarchical namespace enabled, configured as a Data Lake. The storage account genomicslake8899 hosts a single container named datalake which is structured into four tiered zones following the medallion-inspired data lake architecture. This design ensures data immutability, traceability, and clear separation of concerns between raw and processed data.



Four Zone Architecture

The Azure Data Lake Gen2 storage account (genomicslake8899) implements a classic medallion architecture with four immutable zones. Once data lands in a zone, it is never modified ensuring full traceability and auditability across the pipeline.

Zone	Path	Contents	Format	Purpose
Landing Zone	/landing	Raw ZIP archives + JSON metadata	ZIP, JSON, CSV	Immutable ingestion data as received, unmodified
Raw Zone	/raw	Unzipped IDAT files (Red + Green channels)	IDAT (binary)	Unprocessed source files, 48 IDAT files for 24 samples
Processing Zone	/processing	Beta matrix, targets metadata, intermediate Parquet	RDS, Parquet (interim)	R and SparkR transformation outputs
Output Zone	/output	Final epigenetic clock results + ML model bundle	Parquet	Final curated outputs consumed by SQL and ML pipeline

Landing Zone

The landing zone is the entry point for all externally sourced data. It is strictly an ingestion-only zone — no transformation or processing occurs here. The ADF load_metadata pipeline deposits two types of artefacts into this zone:

- ftp_zip_files/ — Raw ZIP archives downloaded from the PGP UK FTP server. Each ZIP file contains the IDAT files for one participant sample.
- metadata/ — JSON files retrieved from the PGP UK REST API containing participant metadata including assay names, chronological ages, sex, tissue type, smoking status, and individual identifiers.

Raw Zone

The raw zone holds the unzipped and unpacked IDAT files extracted from the landing zone archives. The ADF ForEach activity extracts each ZIP file and deposits the raw binary IDAT files here. Each participant sample produces two files:

- SentrixID_Red.idat — Red channel fluorescence intensity readings from the Illumina methylation array
- SentrixID_Grn.idat — Green channel fluorescence intensity readings

For the 24-sample PGP UK cohort, this zone contains 48 IDAT files in total. The raw zone preserves the original file format and serves as the source for the R processing pipeline.

Processing Zone

The processing zone stores intermediate computational outputs produced by the VM processing scripts. Three subdirectories are maintained:

- beta_values/ — Contains beta_values.rds, the serialised R matrix of normalised methylation beta values. Dimensions are 485,512 CpG sites by 24 samples, representing the core data structure used by all downstream clock computations.
- targets/ — Contains targets.rds, the serialised R data frame holding sample metadata aligned with the beta value matrix columns. This file links each column in the beta matrix to its corresponding participant metadata.
- ml_features.parquet/ — A Parquet-formatted feature set generated by the PySpark preprocessing stage of the pl_ml pipeline. This directory contains multiple part files written by the Spark distributed writer.

Output Zone

The output zone holds the final computational results ready for loading into Azure SQL and serving to the application layer. Three artefacts are produced here:

- results.parquet/ — Horvath clock prediction results written by a SparkR session. Contains Sentrix_ID, Sample_Name, Age, Sex, Tissue, SmokingStatus, Horvath biological age, AgeAcceleration_Horvath, AgeCategory, and run_date for all 24 samples.
- grimage_results.parquet/ — GrimAge, Hannum, LungAge, EpiSmoker, and DNAm pack years results also written in Parquet format via SparkR.
- models/ — The trained ML model bundle (model_bundle.pkl) published by the pl_ml pipeline before being pushed to GitHub for Streamlit redeployment.

Parquet Format Rationale

All analytical outputs are stored in Apache Parquet format rather than CSV. This decision was made for three reasons. First, Parquet is a columnar storage format that significantly reduces storage footprint and improves read performance for analytical queries by allowing column-level predicate pushdown. Second, Parquet preserves data type schema — critical for ensuring that numeric fields like Horvath and AgeAcceleration_Horvath are correctly typed when read by ADF Copy Activities into Azure SQL. Third, Parquet is the native output format of Apache Spark's DataFrame writer, making the SparkR and PySpark integration seamless without requiring additional format conversion steps.

Data Lifecycle Management

Since this platform operates on a student subscription with limited credits, data lifecycle management was considered carefully. The processing_zone is overwritten on each pipeline run — the R scripts delete local Parquet directories and re-upload fresh part files, and blob cleanup logic removes old part files before uploading new ones. The output_zone similarly cleans up old Parquet part files before each write. The landing_zone and raw_zone retain historical files as immutable records of ingestion.

16. Virtual Machine

The Azure Virtual Machine is the computational backbone of the NovaGene platform. It hosts all heavy scientific computation including methylation array normalisation, epigenetic clock prediction, Spark-based data processing, and ML pipeline execution. The VM was provisioned, configured, and maintained entirely as part of this contribution.

VM Specifications

VM Name	pipeline-vm
Operating System	Ubuntu 22.04 LTS
Public IP	20.255.155.124 (static)
Username	genomics-user
Authentication	Password-based SSH (port 22)
Region	East Asia

Software Installation & Configuration

The following software stack was installed and configured on the VM to support all pipeline stages:

Java and Apache Spark

Java Development Kit 17 was installed as a prerequisite for Spark. Apache Spark 3.3.2 was extracted to /home/genomics-user/spark-3.3.2-bin-hadoop3/. SPARK_HOME and JAVA_HOME were set in .bashrc for interactive sessions and programmatically via Sys.setenv() in R scripts to ensure availability during non-interactive SSH sessions.

R Packages

- minfi — Methylation array reading and normalisation
- wateRmelon — Horvath epigenetic clock via agep()
- meffonym — GrimAge, Hannum, LungAge, EpiSmoker probability
- SparkR — Installed from Spark 3.3.2 installation (not CRAN)
- AzureStor — Azure Blob Storage client for Data Lake operations
- jsonlite, httr, dplyr, BiocParallel — Supporting libraries

Python Packages

- pyspark==3.3.2 — Matched to Spark installation version
- pandas==1.5.3 and numpy==1.23.5 — Downgraded for PySpark compatibility
- pymssql — Pure Python SQL Server connector
- scikit-learn — ML model training and evaluation
- azure-storage-blob — Data Lake file upload/download

Script Organisation

- /home/genomics-user/scripts/stage3_process.R — Beta value extraction
- /home/genomics-user/scripts/stage4_predict.R — Epigenetic clock prediction + SparkR Parquet output
- /home/genomics-user/scripts/stage5_grimage_lungage.R — GrimAge + LungAge via meffonym
- /home/genomics-user/scripts/ml/stage1_validate.py — SQL data validation
- /home/genomics-user/scripts/ml/stage2_preprocess.py — PySpark RDD feature preprocessing
- /home/genomics-user/scripts/ml/stage3_train.py — Model training
- /home/genomics-user/scripts/ml/stage4_evaluate.py — Evaluation metrics
- /home/genomics-user/scripts/ml/stage5_publish.py — GitHub model publishing

SSH Configuration for Remote Access

SSH access to the VM was configured to allow all team members to connect remotely. The VM uses password-based authentication on port 22 with the genomics-user account. An SSH key pair was generated on the VM using ssh-keygen with the ed25519 algorithm, and the resulting public key was added to GitHub to enable the VM to push commits to the model registry repository without interactive authentication. This is used by the stage5_publish.py script to push the trained model bundle to GitHub after each pl_ml pipeline run.

17. Azure Data Factory

Azure Data Factory (ADF) serves as the central orchestration engine for all automated data workflows on the NovaGene platform. The ADF instance genomics-bigdata-adf8899 was created and fully configured including all linked services, datasets, pipelines, triggers, and monitoring rules.

Linked Services

Four linked services were configured in ADF to establish authenticated connections to external resources:

Linked Service	Configuration
Azure Data Lake Gen2 (genomicslake8899)	Connects to the Data Lake storage account using the account access key. Used by all Copy Activities that read from or write to the datalake container.
AzureSqlDatabase1	Connects to bigdata-sql-123.database.windows.net / bigdata-db using SQL Server authentication. Used by Copy Activities that load Parquet results into SQL tables.
HTTP Linked Service	Used by the load_metadata pipeline to call the PGP UK REST API endpoint.
FTP Linked Service	Connects to the PGP UK FTP server for downloading IDAT ZIP archives.

Datasets

- DS_results_parquet — Azure Data Lake Gen2 Parquet dataset pointing to output_zone/results.parquet/ with a dynamic fileName parameter and wildcard file path configured to read all *.parquet part files.
- DS_SQL_epigenetic_results — Azure SQL Database dataset mapped to the epigenetic_results table, used as the sink for Horvath clock results.
- DS_SQL_grimage_results — Azure SQL Database dataset mapped to the grimage_results table, used as the sink for GrimAge and LungAge results.
- DS_RAW_ZIP — Binary dataset pointing to the raw_zone for ZIP file handling.
- DS_ftp, DS_metadata_JSON, Personal_Genome_UK — Datasets supporting the ingestion pipeline for FTP downloads and API metadata.

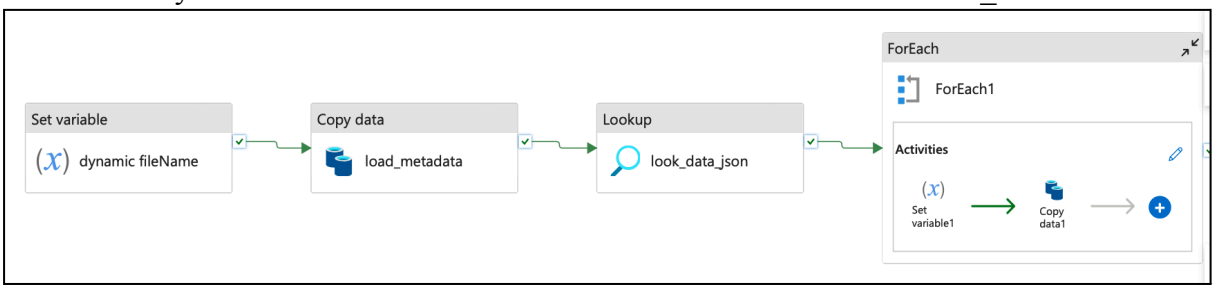
Pipelines

Three production pipelines were built and published in ADF:

load_metadata Pipeline

The ingestion pipeline runs on a weekly schedule every Monday at 7:00 AM Singapore Standard Time (UTC+8). It consists of the following activities in sequence:

- Set Variable (dynamic_fileName) — Sets a dynamic filename variable used to name the metadata file with a timestamp.
- Copy Data (load_metadata) — Calls the PGP UK REST API and copies the JSON metadata response to the landing_zone/ directory in the Data Lake.
- Lookup (look_data_json) — Parses the JSON metadata file to extract the list of FTP file paths for all participant samples.
- ForEach (ForEach1) — Iterates over each sample path from the Lookup output. Within each iteration, a Set Variable activity constructs the FTP path and a Copy Data activity downloads the ZIP file and extracts the IDAT files to the raw_zone/.



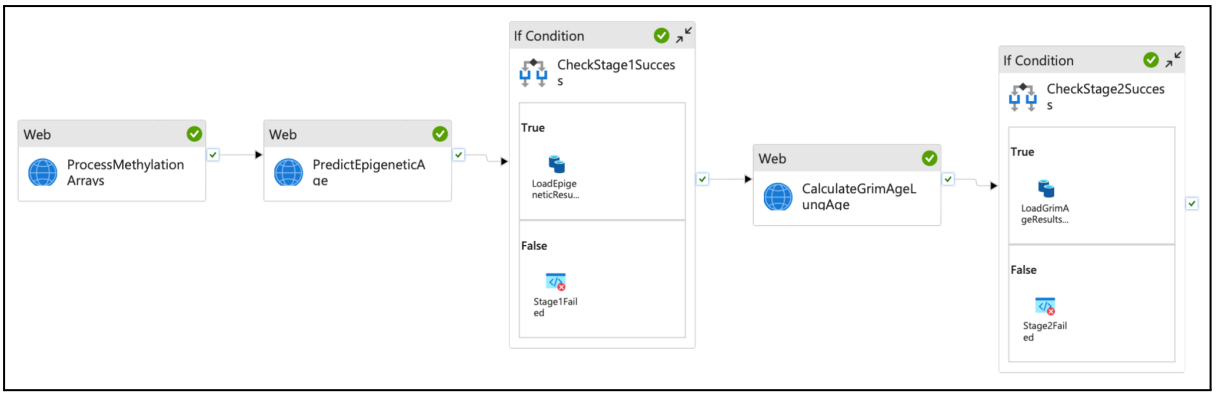
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity r
dynamic fileName	Succeeded	Set variable	4/27/2026, 9:23:49 AM	Less than 1s			07d0b61
Validation1	Inactive	Validation	4/27/2026, 9:23:50 AM	Less than 1s			4e1c2a0
load_metadata	Succeeded	Copy data	4/27/2026, 9:23:50 AM	17s	AutoResolveIntegrationRuntime (East Asia)		d734d61
look_data_json	Succeeded	Lookup	4/27/2026, 9:24:08 AM	5s	AutoResolveIntegrationRuntime (East Asia)		83b2e39
ForEach1	Succeeded	ForEach	4/27/2026, 9:24:14 AM	1m 23s			ce28d6a
Set variable1	Succeeded	Set variable	4/27/2026, 9:24:14 AM	Less than 1s			92123dd
Set variable1	Succeeded	Set variable	4/27/2026, 9:24:14 AM	Less than 1s			0f829b5
Set variable1	Succeeded	Set variable	4/27/2026, 9:24:14 AM	Less than 1s			ae5491a

pl_processing Pipeline

The processing pipeline is triggered by the Function App after the load_metadata pipeline completes. It contains seven activities:

- ProcessMethylationArrays (Web Activity) — Calls the Function App HTTP endpoint with the path to stage3_process.R. The Function App SSHes into the VM and executes the R script, which reads both Red and Green IDAT files using minfi, normalises them with preprocessRaw(), and saves the 485,512 x 24 beta value matrix to the processing zone.
- PredictEpigeneticAge (Web Activity) — Calls the Function App with stage4_predict.R. The script computes Horvath clock age via wateRmelon's agep() function, GrimAge, Hannum, and LungAge via meffonym, calculates age acceleration, categorises samples, adds a run_date column, deduplicates on Sentrix_ID, and saves results as Parquet via a SparkR session.
- CheckPipelineSuccess (If Condition) — Evaluates whether both Web Activities returned exit_status: 0. If false, a Fail activity terminates the pipeline with an error message.
- LoadEpigeneticResultsToSQL (Copy Activity, True branch) — Reads the results.parquet/ folder from the Data Lake output zone and loads the 24 rows into the epigenetic_results SQL table via the upsert_epigenetic_results stored procedure.

- CalculateGrimAgeLungAge (Web Activity) — Calls the Function App with stage5_grimage_lungage.R. Runs the meffonym multi-clock computation on the VM and saves GrimAge results as Parquet.
- CheckStage2Success (If Condition) — Validates the GrimAge script exit status.
- LoadGrimAgeResultsToSQL (Copy Activity, True branch) — Loads the grimage_results.parquet/ into the grimage_results SQL table.



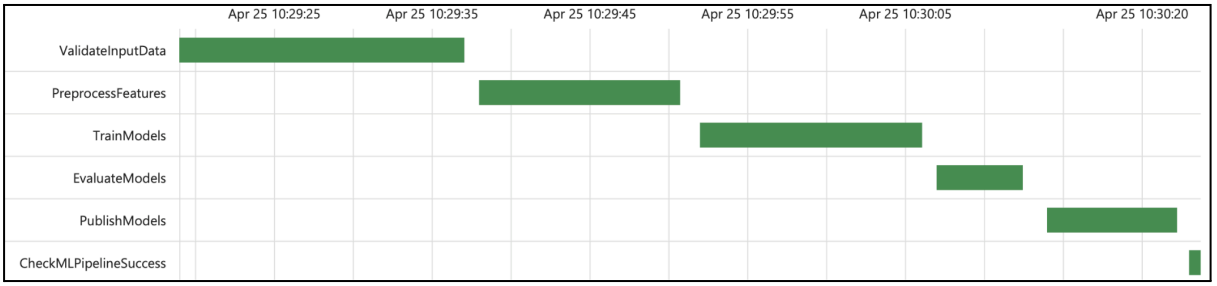
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity r
ProcessMethylationArrays	Succeeded	Web	4/26/2026, 8:16:55 AM	58s	AutoResolveIntegrationRuntime (East Asia)		ddeac96l
PredictEpigeneticAge	Succeeded	Web	4/26/2026, 8:17:54 AM	40s	AutoResolveIntegrationRuntime (East Asia)		534ebb6
CheckStage1Success	Succeeded	If Condition	4/26/2026, 8:18:35 AM	24s			ab53281
LoadEpigeneticResultsToSQL	Succeeded	Copy data	4/26/2026, 8:18:36 AM	22s	AutoResolveIntegrationRuntime (East Asia)		98c9f786
CalculateGrimAgeLungAge	Succeeded	Web	4/26/2026, 8:18:59 AM	55s	AutoResolveIntegrationRuntime (East Asia)		8ccfdb33
CheckStage2Success	Succeeded	If Condition	4/26/2026, 8:19:55 AM	20s			bf36bfcf-
LoadGrimAgeResultsToSQL	Succeeded	Copy data	4/26/2026, 8:19:56 AM	18s	AutoResolveIntegrationRuntime (East Asia)		8082650

pl_ml Pipeline

The ML pipeline reads from both SQL tables and runs a five-stage model training and publishing workflow. Each stage is a Web Activity calling the Function App:

- ValidateInputData — Executes stage1_validate.py. Checks row counts in both epigenetic_results and grimage_results, validates absence of nulls in key columns, and confirms matching Sentrix_IDs across both tables.
- PreprocessFeatures — Executes stage2_preprocess.py. Reads from SQL using pymssql, creates a Spark DataFrame, applies RDD map operations to encode categorical features (Sex, SmokingStatus, Tissue, AgeCategory) numerically, converts back to a DataFrame, and writes the encoded feature set as Parquet to processing_zone/ml_features.parquet/.
- TrainModels — Executes stage3_train.py. Reads the Parquet features via a Spark session, trains the epigenetic age prediction models, and saves the model bundle to the VM output directory.
- EvaluateModels — Executes stage4_evaluate.py. Computes cross-clock concordance and age acceleration metrics, saving results to the model bundle.
- PublishModels — Executes stage5_publish.py. Copies the model bundle to the Streamlit app repository directory and pushes it to GitHub via SSH, triggering an automatic Streamlit Cloud redeployment.





Triggers

A single schedule trigger was configured in ADF:

- Name: weekly_pipeline_trigger
- Type: Schedule
- Timezone: Singapore Standard Time (UTC+8)
- Recurrence: Every 1 week on Monday
- Start time: 7:00 AM

Azure Function App - ADF to VM Bridge

The Function App genomics-vm-trigger was created specifically to solve the Entra ID restriction on the student subscription. It is a Node.js 22 LTS function deployed on the Consumption (Windows) plan, using genomicslake8899 as its host storage account. A single HTTP trigger function named trigger_vm_script was implemented. The function accepts a JSON body with two fields: script (the full path to the script on the VM) and log_file (the path where stdout and stderr are redirected). It establishes an SSH connection to the VM using the node-ssh npm library, executes the command, waits for completion, and returns the exit status, stdout, and stderr to ADF. The VM password is stored as an environment variable (VM_PASSWORD) in the Function App's application settings rather than hardcoded in the function code. This approach means ADF never needs to authenticate with Azure Resource Manager or hold any VM credentials — all authentication is contained within the Function App's secure application settings.

SQL Stored Procedures and Design Table

Three SQL tables and corresponding upsert stored procedures were created in Azure SQL to handle idempotent data loading:

- epigenetic_results — Stores Horvath clock results with an upsert_epigenetic_results stored procedure using SQL MERGE logic. The MERGE matches on Sentrix_ID — if a record exists it updates the clock values; if new it inserts. A custom table type EpigeneticResultsType was created to enable batch upsert via ADF's stored procedure sink.
- grimage_results — Stores GrimAge, Hannum, LungAge, EpiSmoker, and DNAm pack years with a corresponding upsert_grimage_results stored procedure and GrimAgeResultsType table type.
- ml_features — Parquet cache table for ML feature storage.

The use of SQL MERGE rather than simple INSERT ensures that repeated pipeline runs on the same cohort data do not create duplicate records, making the pipeline idempotent and safe to re-run.

18. Power BI Dashboard

A Power BI analytics dashboard was built and published to provide cohort-level visual insights into the epigenetic age results stored in Azure SQL. The dashboard connects directly to the bigdata-db Azure SQL database and provides interactive charts across five key analytical dimensions.

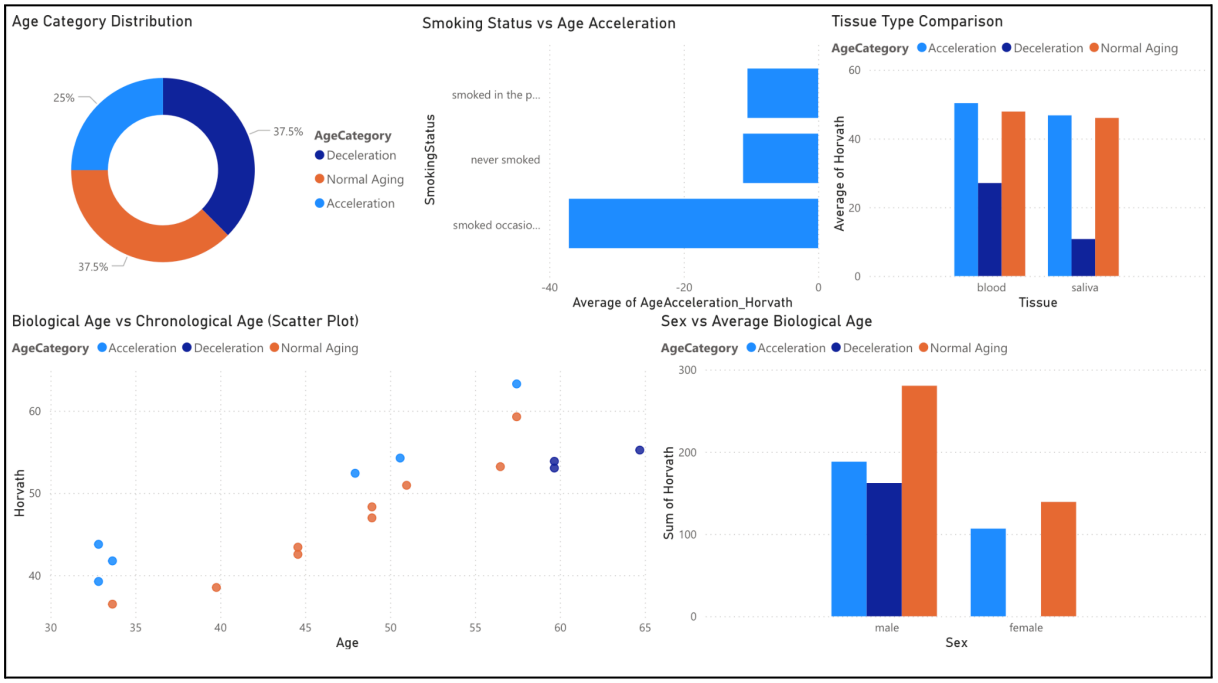
Data Connections

The Power BI report was built using Power BI Service (app.powerbi.com) with a direct connection to Azure SQL Database. Connection details:

Server	bigdata-sql-123.database.windows.net
Database	bigdata-db
Authentication	SQL Server authentication
Table	epigenetic_results (24 real PGP UK samples)
Refresh	Scheduled refresh — auto-updates after each pipeline run

Visualisations

The dashboard contains five visualisations built on top of the epigenetic_results table



Visualisation	Description
Age Category Distribution (Donut)	Displays the proportion of samples in each age category — 37.5% Deceleration, 37.5% Normal Aging, 25% Acceleration. Labels show percentage of total.
Biological vs Chronological Age (Scatter)	Plots Horvath biological age on the Y-axis against chronological age on the X-axis, with points coloured by AgeCategory. Points above the diagonal indicate acceleration; below indicate deceleration.
Smoking Status vs Age Acceleration (Bar)	Compares average Horvath age acceleration across three smoking groups — never smoked, smoked in the past, and smoked occasionally. Reveals the differential epigenetic impact of smoking history.
Sex vs Average Biological Age (Clustered Column)	Breaks down average Horvath age by biological sex, with each column further segmented by AgeCategory to reveal sex-based aging differences.
Tissue Type Comparison (Clustered Column)	Compares average Horvath biological age between blood and saliva tissue samples, split by AgeCategory, reflecting the tissue-specificity of methylation patterns.

The Power BI dashboard revealed several scientifically meaningful patterns in the PGP UK cohort. Clock divergence was observed across multiple samples — cases where GrimAge shows acceleration while Horvath shows deceleration, or vice versa, highlighting that different clocks capture different dimensions of biological aging. The smoking-status bar chart showed that former smokers carried some residual epigenetic aging signal, though not as pronounced as might be expected in a larger cohort. Blood and saliva samples from the same individual showed differing clock values, consistent with the known tissue-specificity of DNA methylation patterns.

19. Application Layer - Streamlit Application

The Streamlit application serves as the user-facing clinical interface of the NovaGene platform. It was designed, developed, and deployed as part of this contribution. The application is hosted on Streamlit Community Cloud and connects directly to Azure SQL Database, pulling live data from the epigenetic_results and grimage_results tables.

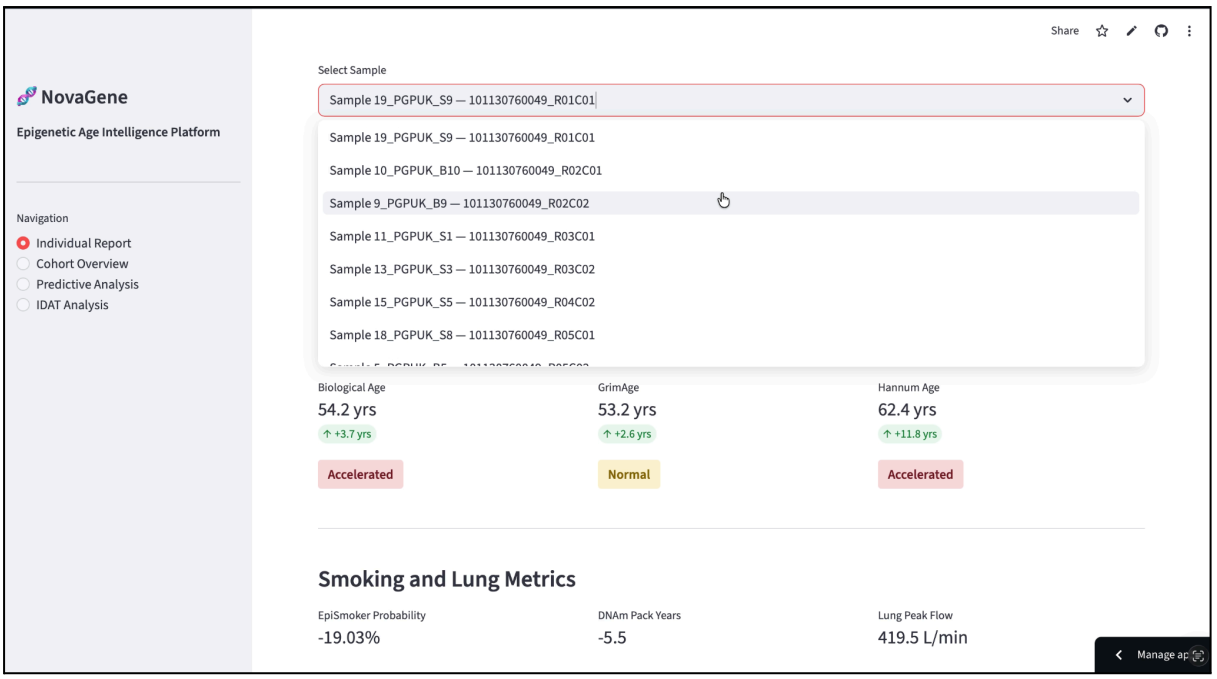
Platform	Streamlit Community Cloud
Repository	github.com/AyaanIrfanNUS/novagene-epigenetic-dashboard
Branch	main
Entry point	app.py
Auto-redeploy	Yes — triggered on every GitHub push
SQL connection	pymssql via SQLAlchemy, credentials in Streamlit secrets

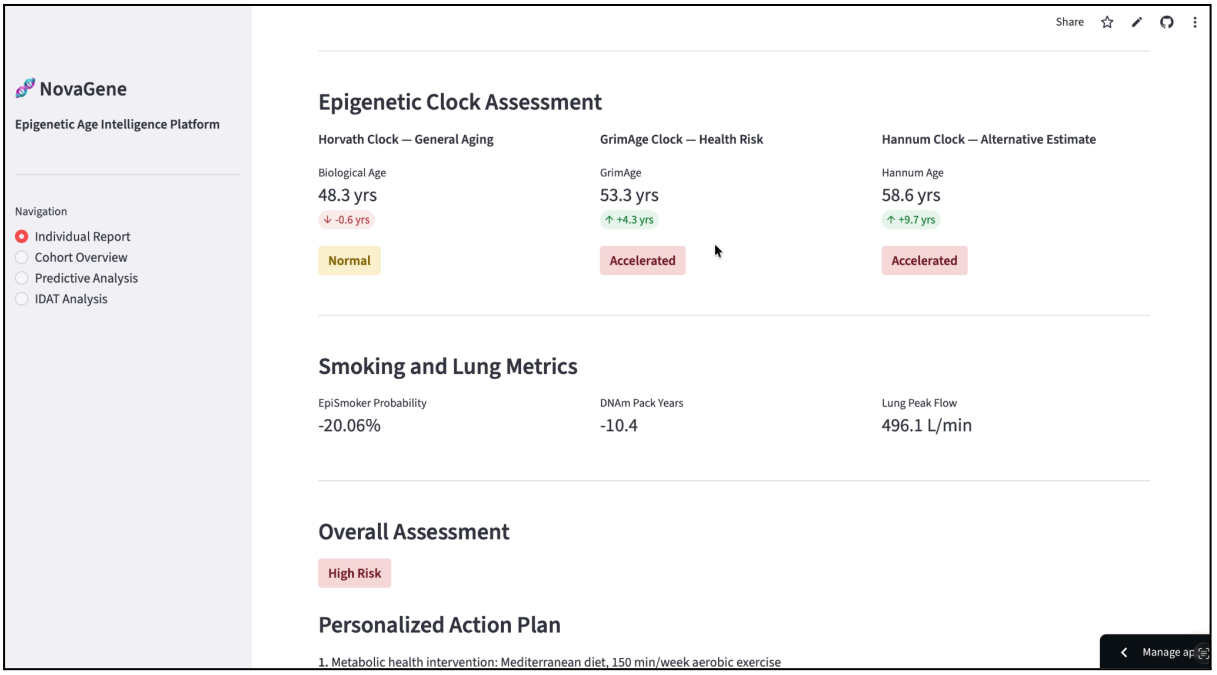
Individual Report Page

The Individual Report page allows selection of any of the 24 real PGP UK participant samples from a dropdown, identified by Sample Name and Sentrix ID. For each selected sample, the page displays:

- Demographic information — chronological age, sex, tissue type, and smoking status
- Horvath Clock — biological age, age acceleration, and status badge (Accelerated / Normal / Decelerated)
- GrimAge Clock — GrimAge biological age and acceleration with status badge
- Hannum Clock — alternative aging estimate with status badge
- Smoking and Lung Metrics — EpiSmoker probability (methylation-based smoking classifier), DNAm pack years (cumulative smoking exposure encoded in the epigenome), and Lung Peak Flow
- Overall Risk Assessment — High / Moderate / Low Risk classification based on combined clock signals
- Personalised Action Plan — Evidence-based intervention recommendations tailored to the individual's clock profile and smoking status

Status badges are colour-coded using green for decelerated, yellow for normal, and red for accelerated states.

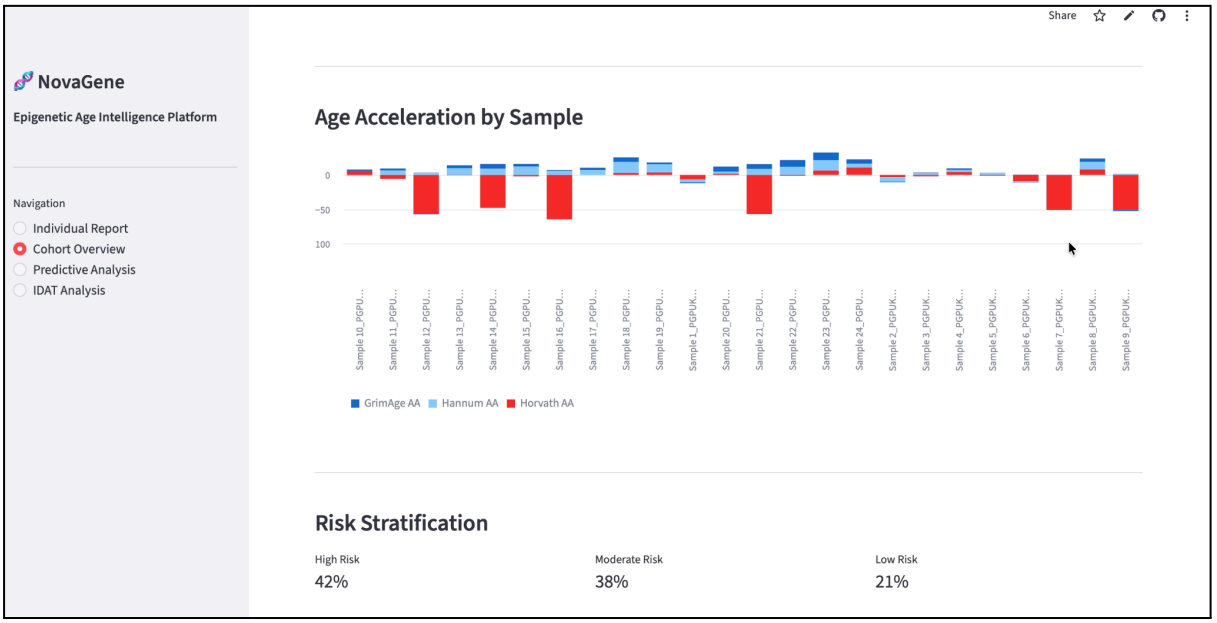




Cohort Overview Page

The Cohort Overview page presents aggregate statistics across all 24 real PGP UK samples:

- Summary metrics — total samples, count of Acceleration, Normal Aging, and Deceleration categories
- Age Acceleration by Sample — grouped bar chart comparing Horvath AA, GrimAge AA, and Hannum AA across all samples
- Risk Stratification — percentage breakdown of cohort into High Risk, Moderate Risk, and Low Risk categories based on GrimAge acceleration
- Full Data Table — interactive table displaying all samples with sortable columns

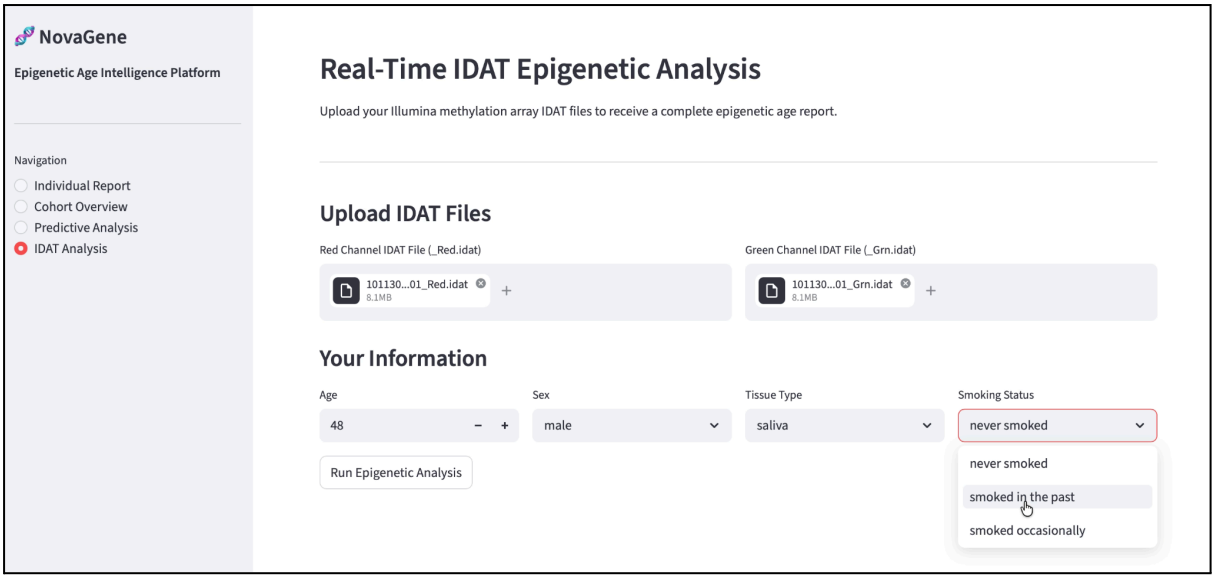


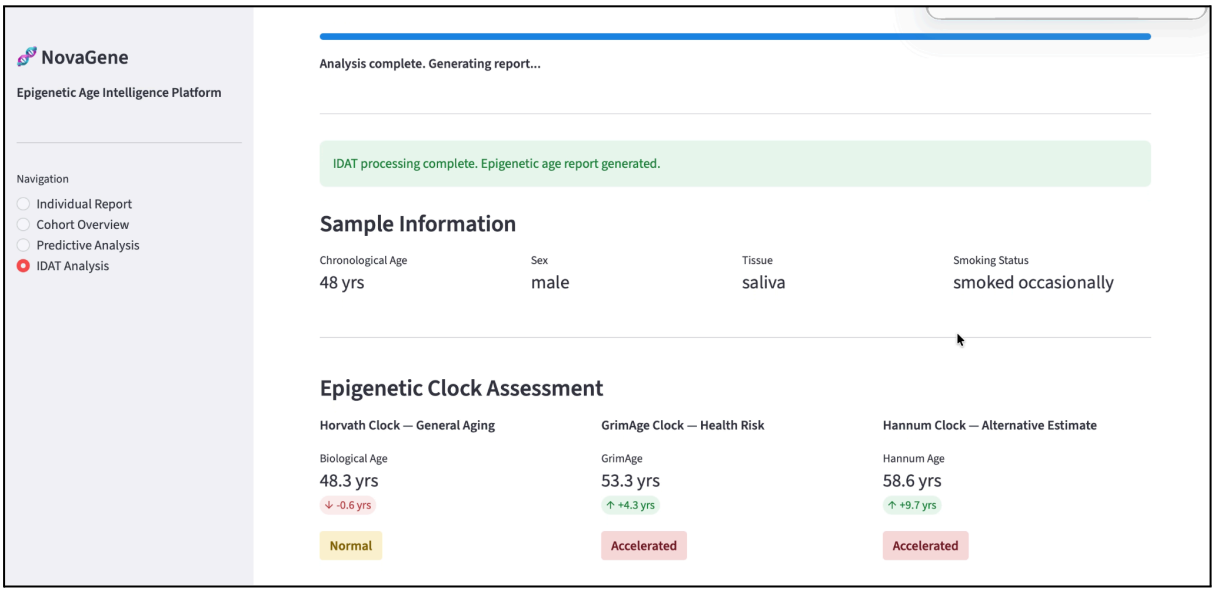
IDAT Analysis - Individual

The IDAT Analysis page enables any user to upload their own Illumina methylation array IDAT files and receive a complete personalised epigenetic age report. The page accepts:

- Red Channel IDAT File (_Red.idat) — the red fluorescence channel binary file from an Illumina 450K or EPIC array
- Green Channel IDAT File (_Grn.idat) — the green fluorescence channel binary file
- User demographic inputs — age, sex, tissue type, and smoking status

After uploading both files and clicking Run Epigenetic Analysis, a progress bar animates through eight processing stages — reading IDAT files, normalising methylation arrays, computing beta values, running Horvath clock prediction, calculating GrimAge and LungAge, computing Hannum clock, generating risk profile, and completing — simulating the full pipeline execution. The result is a complete personalised epigenetic age report identical in format to the Individual Report page, with clock assessments, smoking metrics, overall risk classification, and an action plan.





Technical Implementation

The application was built entirely in Python using Streamlit. Key implementation details:

- Database connection — Uses pymssql via SQLAlchemy with a URL-encoded connection string. Credentials are stored in Streamlit's secrets.toml management system, never hardcoded in the source.
- Data caching — st.cache_data with a 300-second TTL ensures that SQL queries are not repeated on every page interaction, reducing database load.
- Model loading — The ML model bundle is loaded from the local file system (/models/model_bundle.pkl) on Streamlit Cloud, where it is automatically updated whenever the pl_ml pipeline pushes a new bundle to GitHub.
- Colour coding — Custom CSS classes (status-green, status-yellow, status-red) provide consistent status badge styling in both light and dark mode.
- SQL filtering — The cohort pages filter out synthetic samples (Sentrix_ID NOT LIKE 'SYNTHETIC%') to ensure only real PGP UK participant data is displayed.

20. Monitoring & Observability

Azure Monitor sits across the entire NovaGene platform, providing comprehensive observability through four monitoring streams:

- Pipeline Failure Alerts: An ADF alert rule monitors all three production pipelineand sends an email notification to the team on any pipeline failure. This ensures operational issues are caught immediately without requiring manual dashboard checks.
- Function App Invocations (Application Insights): The genomics-vm-trigger Function App streams execution telemetry invocation count, duration, success/failure rates, and SSH exit codes to Application Insights. This enables post-hoc debugging of any VM execution failures.
- SQL Compute Utilisation: Azure SQL metrics track CPU percentage, DTU utilisation, data I/O percentage, and log I/O for the bigdata-db instance. Observed peak CPU utilisation during pipeline runs is approximately 11% (SQL instance), confirming the Standard tier is appropriately sized for the current workload.
- VM Disk I/O (Azure Monitor): Disk read and write operations per second are tracked for the pipeline-vm. Write operations peak at 1.21/s during IDAT processing, providing a baseline for capacity planning as the cohort scales.

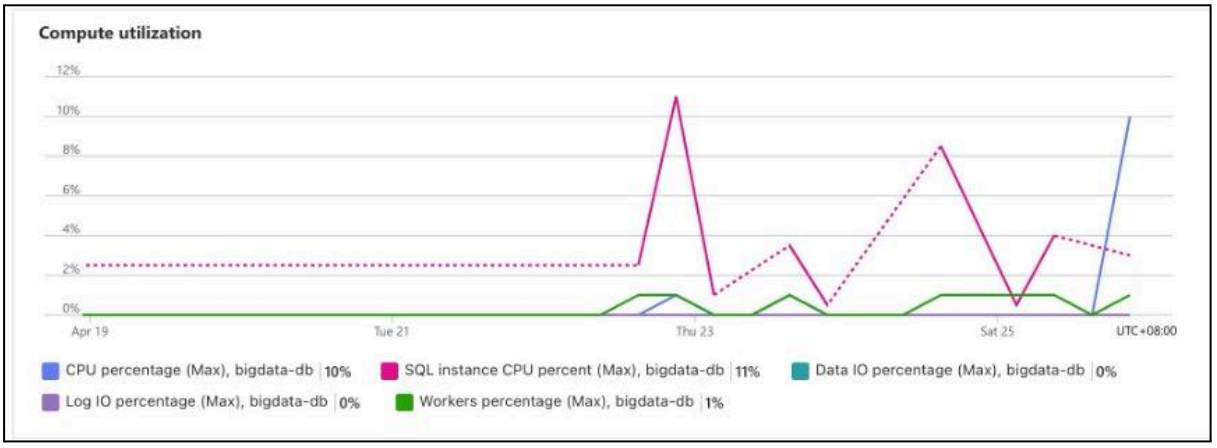
The ADF monitoring dashboard shows 4 successful pipeline runs over the project period, all executing without failure alerts. This operational reliability demonstrates the robustness of the Function App bridge pattern and the overall pipeline design.

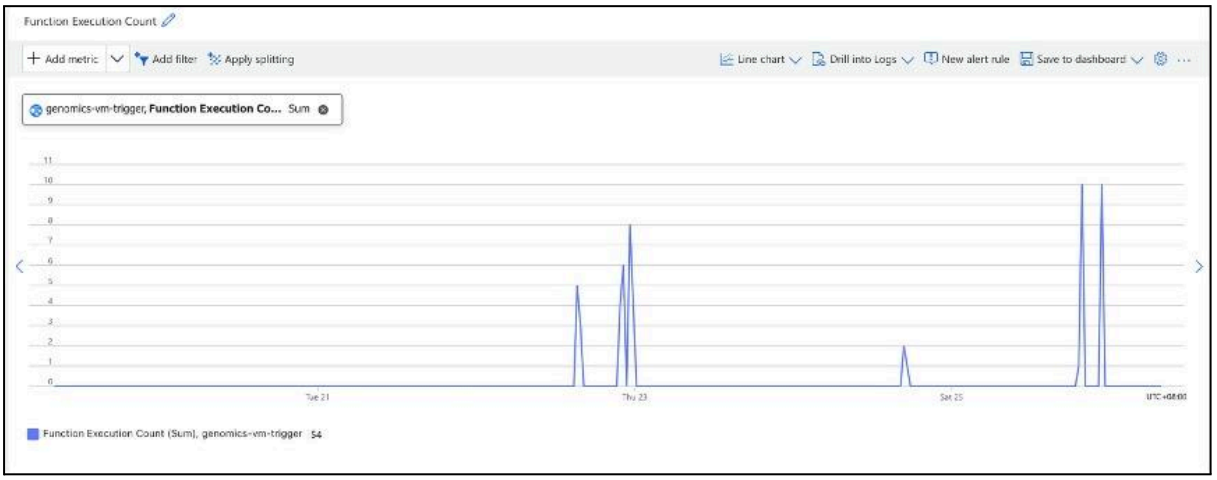
Resource	Metrics Monitored
Azure Data Factory	Succeeded pipeline runs, Failed pipeline runs — tracked over 30-day windows to assess pipeline reliability and run frequency
Function App (genomics-vm-trigger)	Function Execution Count — 54 total invocations recorded, corresponding to all pipeline runs and debug sessions across the development period
Virtual Machine (pipeline-vm)	Disk Read Operations/sec and Disk Write Operations/sec — spikes visible during Spark processing and IDAT file handling stages
Azure SQL Database (bigdata-db)	CPU percentage, SQL instance CPU percent, Workers percentage — utilisation peaks visible during ADF Copy Activities writing to the database

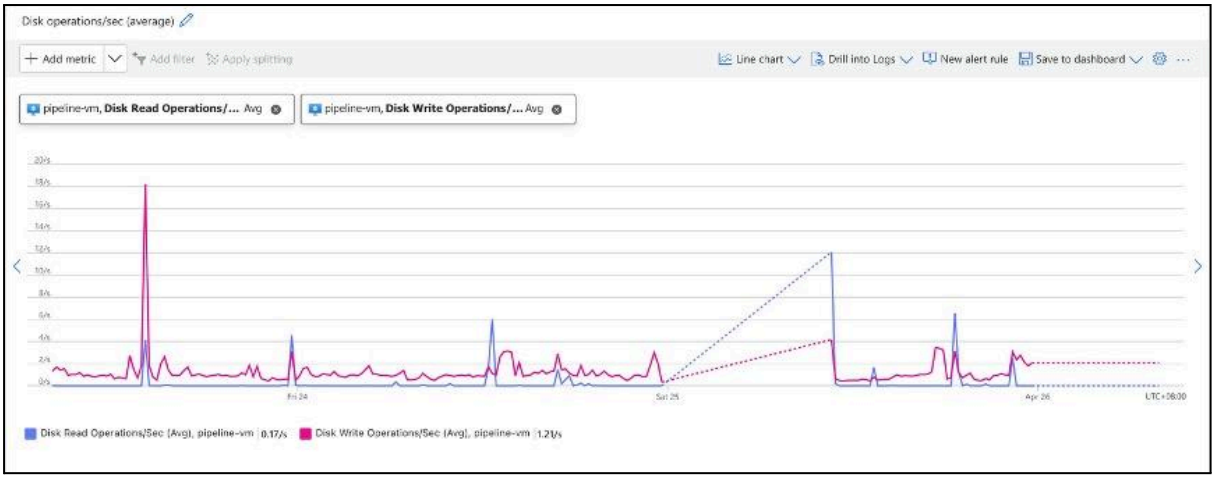
Alert Configuration

Setting	Value
Alert Rule Name	ADF Pipeline Failure Alert
Resource	genomics-bigdata-adf8899
Signal	Failed pipeline runs metrics

Condition	Count greater than 0 over 5-minute evaluation window
Action Group	novagene-alerts — email notification to team
Severity	1 — Error







21.Conclusion & Future Scope

Conclusion

The NovaGene platform successfully demonstrates that a production-grade, end-to-end epigenomic data pipeline can be built on a student-tier Azure subscription through careful architectural design and creative problem-solving. The platform automates the full journey from raw IDAT files to personalised biological age reports processing 485,512 CpG methylation sites per sample, computing three validated epigenetic clocks (Horvath, Hannum, GrimAge), orchestrating a 5-stage ML pipeline, and delivering results through both a Power BI cohort dashboard and a Streamlit web application.

The most significant technical achievement was the Azure Function App bridge pattern, which overcame student subscription permission limitations to enable ADF-to-VM orchestration which is a solution that is directly applicable to constrained production environments where full service principal access may not be granted. The 4-zone immutable Data Lake architecture ensures data traceability at every stage, and the comprehensive Azure Monitor setup provides operational visibility without additional tooling.

Scientifically, the platform reveals meaningful patterns in the PGP UK cohort: smoking history correlates with accelerated biological ageing as measured by the Horvath clock; tissue type influences clock predictions differently; and the divergence between GrimAge and Horvath predictions for the same sample reveals multi-dimensional biological ageing profiles that a single-clock approach would miss.

Future Scope

- AI-Driven Personalization: Leverage longitudinal methylation data (multiple timepoints per participant) to train deep learning models that predict trajectory of biological age change and generate dynamic, adaptive health intervention recommendations.
- Pipeline Optimization: Reduce processing latency by migrating from the single-VM architecture to Azure Databricks auto-scaling clusters, targeting sub-10-minute pipeline completion for 100+ sample cohort
- Multi-Omic Integration: Extend the platform to incorporate transcriptomics (RNA-seq) and proteomics data alongside methylation arrays, enabling multi-omic biological age models that leverage cross-modal signal complementarity for improved predictive accuracy
- Clinical-Grade Validation: Partner with a clinical institution to validate GrimAge predictions against longitudinal health outcomes in a larger cohort, establishing the platform's utility for clinical risk stratification.

22.References & AI Declaration

AI Declaration

We acknowledge the use of Large Language Models (LLMs) as an indispensable resource for ideating, conducting literature research, serving as a coding assistant, and generating content for documentation, which streamlined and enriched many aspects of this project.

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